# Functions - Maxima, Minima and Monotonicity: Part 1

Course: Pre-Calculus-I

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# Outline

1 Increasing and Decreasing Functions

2 Maximum and Minimum Values

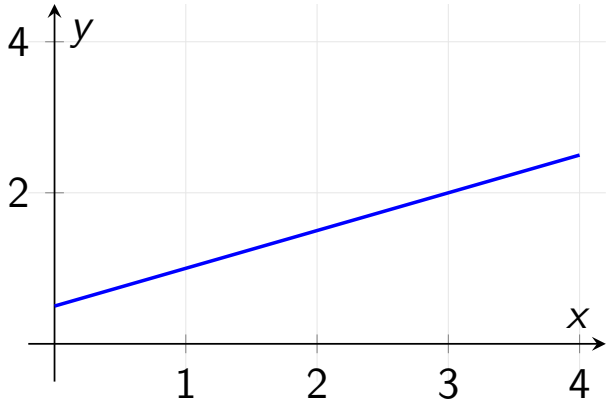
3 Max/Min and Monotonicity by Function Type

# Increasing and Decreasing: Intuition

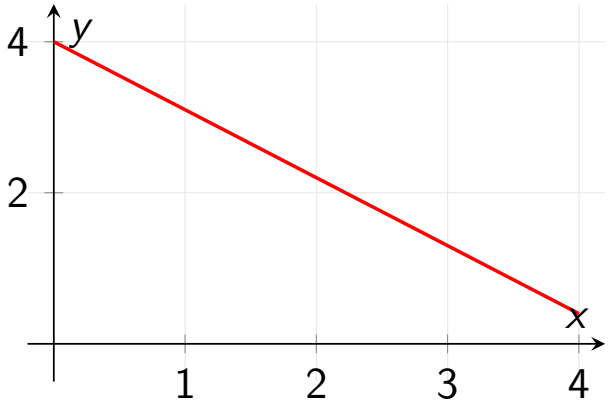
**Informally:**

- A function is increasing if its graph goes *up* as we move left to right.
- A function is decreasing if its graph goes *down* as we move left to right.

Increasing




Decreasing



# Formal Definition

Let $f$ be defined on an interval $I$.

## Increasing

$f$ is increasing on $I$ if for all $x_1, x_2 \in I$:

$$x_1 < x_2 \implies f(x_1) < f(x_2)$$

## Decreasing

$f$ is decreasing on $I$ if for all $x_1, x_2 \in I$:

$$x_1 < x_2 \implies f(x_1) > f(x_2)$$

(If we allow $\leq$ / $\geq$ instead of strict inequalities, we say *non-decreasing* / *non-increasing*.)

# How to Check Increasing / Decreasing

## Method 1: Table of Values

Pick test points $x_1 < x_2 < x_3 < \ldots$ in the interval and compare the corresponding $f(x)$ values.

## Method 2: Algebraic Comparison

For arbitrary $x_1 < x_2$, show $f(x_2) - f(x_1) > 0$ (increasing) or $< 0$ (decreasing).

## Method 3: Graph Inspection

Trace the curve from left to right and see whether it rises or falls.

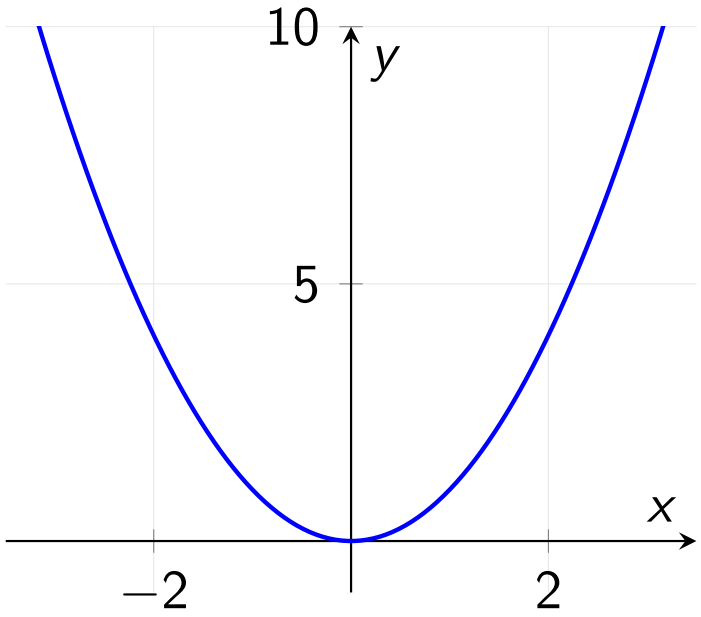
## Note

*For quadratic functions the vertex separates the increasing part from the decreasing part.*

# Example: $f(x) = x^2$ (Graph and Table)

**Table of values:**

| $x$ | $f(x) = x^2$ |
|---|---|
| $-3$ | 9 |
| $-2$ | 4 |
| $-1$ | 1 |
| 0 | 0 |
| 1 | 1 |
| 2 | 4 |
| 3 | 9 |




**Observations from the table:**

- As $x$ goes from $-3$ to 0: values fall $9 \to 4 \to 1 \to 0 \Rightarrow$ decreasing.
- As $x$ goes from 0 to 3: values rise $0 \to 1 \to 4 \to 9 \Rightarrow$ increasing.
- $f(0) = 0$ is the smallest value $\Rightarrow$ absolute minimum.

# Absolute vs. Local Extrema

## Absolute (Global) Extrema

- $f(c)$ is an absolute maximum if $f(c) \geq f(x)$ for all $x$ in the domain.
- $f(c)$ is an absolute minimum if $f(c) \leq f(x)$ for all $x$ in the domain.

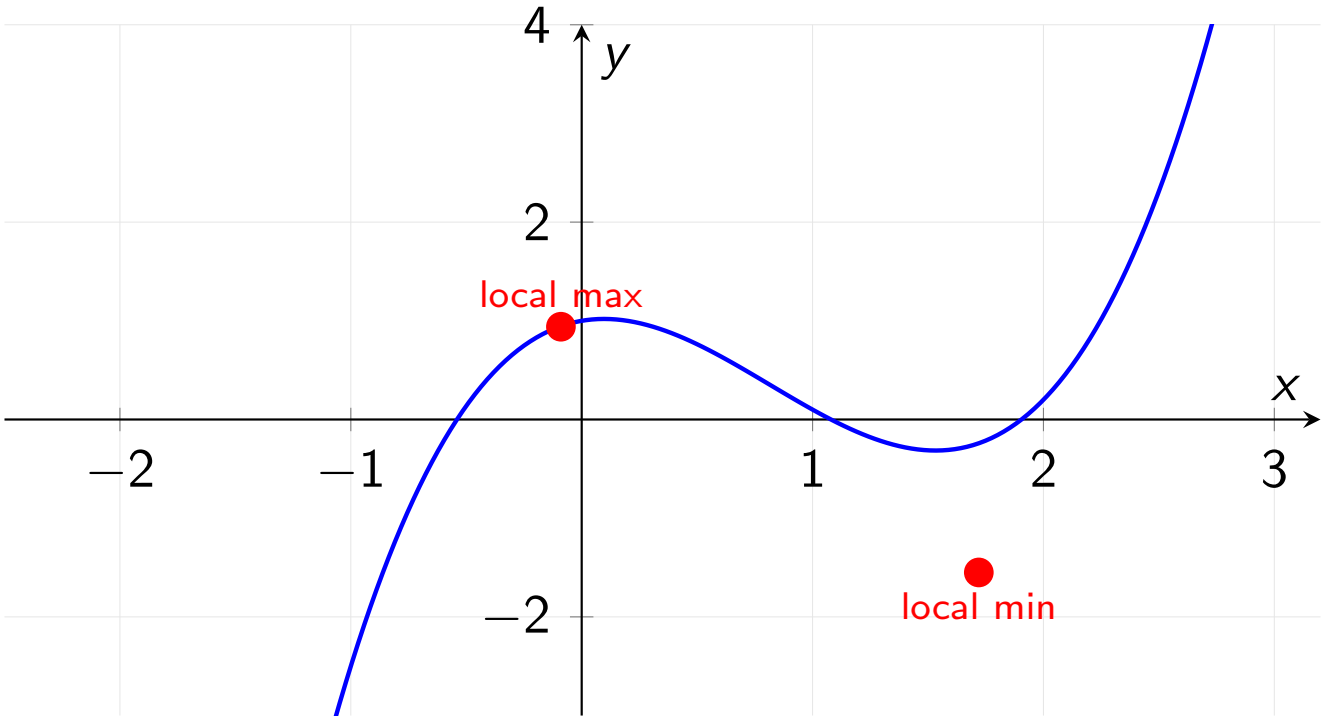
## Local (Relative) Extrema

- $f(c)$ is a local maximum if $f(c) \geq f(x)$ for all $x$ *near* $c$.
- $f(c)$ is a local minimum if $f(c) \leq f(x)$ for all $x$ *near* $c$.

## Note

*Every absolute extremum is also a local extremum (when interior), but not conversely.*

# Visualising Extrema




A local maximum is a *peak*; a local minimum is a *valley* of the graph.

# How to Find Maximum and Minimum Values

## Method 1: Graph / Table Inspection

Plot or tabulate the function and read off the highest and lowest points.

## Method 2: Completing the Square (Quadratics)

Rewrite $f(x) = ax^2 + bx + c$ as $f(x) = a(x - h)^2 + k$; the vertex $(h, k)$ gives the extremum.

## Method 3: Bounded Ranges

Use known ranges of standard functions (e.g. $\sin x, \cos x \in [-1, 1]$, $|x| \geq 0$, $a^x > 0$).

## Method 4: Closed-Interval Comparison

On $[a, b]$, evaluate $f$ at the endpoints and at any interior peak/valley and compare.

# Peak / Valley Test (No Calculus)

At a point $c$, examine the values of $f$ just to the *left* and just to the *right* of $c$.

| **Pattern around $c$** | **Shape** | **Conclusion** |
|---|---|---|
| $f$ rises then falls | peak $\wedge$ | local maximum at $c$ |
| $f$ falls then rises | valley $\vee$ | local minimum at $c$ |
| $f$ keeps rising | – | neither (no extremum) |
| $f$ keeps falling | – | neither (no extremum) |

**Note**

*On a closed interval, always compare interior peaks/valleys with the endpoint values.*

# Absolute Extrema on a Closed Interval

## Extreme Value Theorem

If $f$ is continuous on a closed interval $[a, b]$, then $f$ attains both an absolute maximum and an absolute minimum on $[a, b]$.

**Closed-Interval Method (no derivatives):**

1. Locate any peaks and valleys inside $(a, b)$ from the graph or table.
2. Evaluate $f$ at those points.
3. Evaluate $f$ at the endpoints $a$ and $b$.
4. The largest value is the absolute maximum; the smallest is the absolute minimum.

## Note

*Endpoints are always candidates on a closed interval.*

# Linear Functions: $f(x) = mx + c$

| **Slope** | **Monotonicity** | **Extrema on $\mathbb{R}$** |
|---|---|---|
| $m > 0$ | increasing on $\mathbb{R}$ | none |
| $m < 0$ | decreasing on $\mathbb{R}$ | none |
| $m = 0$ | constant | every point is both max & min |

**Why?** From $f(x) = mx + c$:

$$f(x_2) - f(x_1) = m(x_2 - x_1).$$

If $m > 0$ and $x_2 > x_1$, then $f(x_2) > f(x_1)$; if $m < 0$, then $f(x_2) < f(x_1)$.

**Example:** $f(x) = -3x + 7$ on $[0, 4]$.

- $m = -3 < 0 \Rightarrow f$ decreasing.
- Absolute maximum at $x = 0$: $f(0) = 7$.
- Absolute minimum at $x = 4$: $f(4) = -5$.

# Quadratic Functions: $f(x) = ax^2 + bx + c$

**Vertex form:** $f(x) = a\left(x + \frac{b}{2a}\right)^2 + k$, vertex at $x_v = -\dfrac{b}{2a}$.

| Leading coefficient | Opens | Vertex is |
|---|---|---|
| $a > 0$ | upward $\smile$ | absolute minimum |
| $a < 0$ | downward $\frown$ | absolute maximum |

**Monotonicity from the vertex:**

- $a > 0$: decreasing on $(-\infty, x_v)$, increasing on $(x_v, \infty)$.
- $a < 0$: increasing on $(-\infty, x_v)$, decreasing on $(x_v, \infty)$.

**Note**

*Since $(x - h)^2 \geq 0$, the expression $a(x - h)^2 + k$ is minimised (or maximised) at $x = h$.*

# Example: Quadratic Function

Let $f(x) = 2x^2 - 8x + 5$.

**Step 1 (completing the square):** $f(x) = 2(x-2)^2 - 3$.

**Step 2:** Vertex at $x = 2$, value $f(2) = -3$. Since $a = 2 > 0$, it is an absolute minimum.

**Step 3:** Monotonicity: Decreasing on $(-\infty, 2)$ and Increasing on $(2, \infty)$.

